

1. Definition:

OpenStack là một nền tảng IaaS gồm nhiều service độc lập. Mỗi service phụ trách một nhóm chức năng riêng.

Nhóm chức năng	Service chính	Vai trò
Identity	Keystone	Xác thực, phân quyền, cấp token
Image	Glance	Quản lý image hệ điều hành
Compute	Nova	Tạo và quản lý máy ảo
Resource tracking	Placement	Theo dõi CPU, RAM, disk và allocation
Networking	Neutron	Quản lý network, subnet, router, port, Floating IP
Block storage	Cinder	Quản lý volume gắn vào VM
Object storage	Swift	Lưu object qua HTTP API
Shared file system	Manila	Cung cấp file share dùng chung
Dashboard	Horizon	Giao diện web
Orchestration	Heat	Tạo nhiều tài nguyên bằng template
Workflow	Mistral	Thực thi workflow nhiều bước
Load balancing	Octavia	Cung cấp Load Balancer as a Service
Key management	Barbican	Quản lý secret, key, certificate
Database hạ tầng	MariaDB	Lưu metadata và trạng thái của các service
API high availability	HAProxy + Keepalived	Cung cấp VIP và chuyển tiếp API

Ngoài ra, hệ thống thường có RabbitMQ để làm message queue, Memcached để cache token và Open vSwitch hoặc OVN để xử lý mạng ảo.

Các service OpenStack giao tiếp thông qua 3 lớp chính: REST API, message queue, database

REST API: User or CLI gọi OpenStack

- Service gọi API của service khác
- Trao đổi qua HTTP/HTTPS
- Dữ liệu thường ở dạng json

- Xác thực bằng token của Keystone
- Ví dụ : user -> Nova API, Nova API -> Glance API, Neutron API, Placement API, Cinder -> Keystone API

Message q:

- Hệ thống trung gian giúp các dịch vụ như Nova, Neutron, Cinder phối hợp hoạt động, trao đổi trạng thái và truyền tải lệnh điều khiển một cái đồng bộ hoặc bất đồng bộ. Message q giúp API ko phải chờ toàn bộ tác vụ hoàn thành

Database:

- Database (MariaDB) lưu: metadata, trạng thái resource, mapping, configuration nghiệp vụ, thông tin instance, network, port, volume, image.
- Glance database lưu metadata image
- Neutron db lưu volume, snapshot và attachment
- Keystone db lưu user, project, role và endpoint
- Placement DB lưu inventory và allocation

Cơ chế	Mục đích	Ví dụ
REST API	Giao tiếp giữa client và service hoặc giữa các service	Nova gọi Neutron
Message queue	Gửi lệnh, RPC và event nội bộ	Nova Conductor gửi lệnh tới Nova Compute
Database	Lưu trạng thái và metadata lâu dài	Nova lưu trạng thái instance

Keystone = IAM:

- Manages Identity, Authentication, Authorization, Token, Project và domain, Role, Service Catalog, Endpoint của các service
- Keystone luôn phải sẵn sàng, ví dụ khi user execute openstack server list -> Client sẽ gửi thông tin đăng nhập đến Keystone -> Keystone xác thực user -> Keystone cung cấp token -> Client lấy service catalog -> Client tìm endpoint của Nova -> Nova check token hoặc thông tin xác thực với middleware Keystone -> Nếu keystone không hoạt động: -> user không lấy được token, service không lấy được verify request, CLI và Horizon không đăng nhập, service to service request fail, gián đoạn administration.

Phân biệt các loại storage

Cần phân biệt rõ ba loại storage chính trong OpenStack:

Glance – Image storage

- Quản lý các image dùng để khởi tạo VM.
- Image thường chứa hệ điều hành và các phần mềm được cài đặt sẵn.
- Ví dụ: Ubuntu image, Rocky Linux image hoặc Windows Server image.

Use case:

- Upload image hệ điều hành.
- Dùng image làm nguồn để tạo VM mới.
- Quản lý các phiên bản image chuẩn của doanh nghiệp.

Cinder – Block storage

- Cung cấp volume dạng block cho VM.
- Volume hoạt động tương tự như một ổ cứng được gắn vào máy ảo.
- Dữ liệu có thể tồn tại ngay cả khi VM bị xóa, tùy theo cấu hình.

Use case:

- Gắn thêm ổ đĩa cho VM.
- Lưu database hoặc dữ liệu ứng dụng.
- Tạo snapshot và backup volume.
- Di chuyển volume giữa các VM khi cần.

Swift – Object storage

- Lưu dữ liệu dưới dạng object trong container.
- Không được gắn vào VM như một ổ đĩa thông thường.
- Thường được truy cập thông qua API HTTP.

Use case:

- Lưu file backup.
- Lưu ảnh, video hoặc tài liệu.

- Lưu log và dữ liệu dung lượng lớn.
- Lưu dữ liệu ít thay đổi nhưng cần khả năng mở rộng cao.

Tóm tắt:

- Glance lưu image để tạo VM.
- Cinder cung cấp ổ đĩa block cho VM.
- Swift lưu file và object thông qua API.

Các tiến trình chính của Nova

nova-api

- Nhận request từ user, Horizon hoặc OpenStack CLI.
- Kiểm tra request và chuyển công việc đến các thành phần khác.
- Là điểm truy cập chính của Nova.

nova-scheduler

- Chọn compute node phù hợp để chạy VM.
- Dựa trên CPU, RAM, disk, availability zone và các filter đã cấu hình.
- Tham khảo thông tin tài nguyên từ Placement.

nova-conductor

- Là lớp trung gian giữa nova-compute và database.
- Thực hiện các tác vụ liên quan đến database thay cho nova-compute.
- Giúp compute node không cần truy cập trực tiếp vào database.

nova-compute

- Chạy trên mỗi compute node.
- Giao tiếp với hypervisor như KVM hoặc libvirt.
- Thực hiện việc tạo, khởi động, dừng, reboot hoặc xóa VM.

Phần 3 – Luồng tạo một VM

Cần nắm được luồng hoạt động từ lúc user gửi yêu cầu đến khi VM chạy thành công.

Luồng cơ bản

1. User gửi yêu cầu tạo VM thông qua Horizon, OpenStack CLI hoặc API.
2. Keystone xác thực user và cấp token.
3. Request được gửi đến Nova API cùng với token xác thực.
4. Nova API kiểm tra thông tin request, project, quota và quyền truy cập.
5. Nova API yêu cầu Glance cung cấp thông tin về image được chọn.
6. Nova API yêu cầu Neutron tạo network port và cấp địa chỉ IP cho VM.
7. Nếu VM sử dụng volume, Nova giao tiếp với Cinder để tạo hoặc gắn volume.
8. Nova Scheduler nhận yêu cầu và tìm compute node phù hợp.
9. Nova Scheduler sử dụng thông tin tài nguyên từ Placement để lựa chọn compute node.
10. Yêu cầu được chuyển đến nova-compute trên node đã chọn.
11. Nova-compute lấy image từ Glance hoặc sử dụng volume từ Cinder.
12. Nova-compute yêu cầu Neutron hoàn thiện network port và kết nối mạng cho VM.
13. Nova-compute giao tiếp với hypervisor để tạo và khởi động VM.
14. Trạng thái của VM được cập nhật trong Nova database.
15. User có thể kiểm tra trạng thái VM thông qua Horizon, CLI hoặc API.

Luồng tổng quát:

User

- Keystone
- Nova API
- Glance / Neutron / Cinder
- Nova Scheduler
- Placement
- Nova Compute
- Hypervisor
- VM chạy thành công

2. Lab Test:

README.md (for later upload to Git)

3. Check ubuntu configuration after installed on VMWare – Virtual Box

```
quantha@openstack-aiio:~$ hostnamectl
Static hostname: openstack-aiio
Icon name: computer-vm
Chassis: vm
Machine ID: c94fd5de202349c887360b54fc8f187d
Boot ID: a80234ff21b74cc68a9a7bd1cf8563a
Virtualization: oracle
Operating System: Ubuntu 24.04.4 LTS
Kernel: Linux 6.8.0-136-generic
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 7month 4w
quantha@openstack-aiio:~$ ip -br addr
lo                UNKNOWN    127.0.0.1/8 ::1/128
enp0s3            UP          10.0.2.15/24 metric 100 fd17:625c:f037:2:a00:27ff:fe13:8182/64 fe80::a00:27ff:fe13:8182/64
enp0s8            DOWN
quantha@openstack-aiio:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
8.8.4.4 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
8.8.8.8 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
quantha@openstack-aiio:~$ lsblk -o NAME,SIZE,FSTYPE,MOUNTPOINTS
NAME        SIZE FSTYPE MOUNTPOINTS
sda          70G
├─sda1       1M
├─sda2       2G ext4   /boot
├─sda3       68G LVM2_member
└─ubuntu--vg-ubuntu--lv 34G ext4   /
sdb          30G
sr0          1024M
quantha@openstack-aiio:~$ egrep -c '(vmx|svm)' /proc/cpuinfo
0
quantha@openstack-aiio:~$ ls -l /dev/kvm
ls: cannot access '/dev/kvm': No such file or directory
quantha@openstack-aiio:~$
```

4. Extend available disk storage

```
quantha@openstack-aiio:~$ df -h /
Filesystem              Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 34G  6.6G  25G  21% /
quantha@openstack-aiio:~$ sudo vgs
[sudo] password for quantha:
VG      #PV #LV #SN Attr   VSize   VFree
ubuntu-vg  1  1  0 wz--n- <68.00g 34.00g
quantha@openstack-aiio:~$ sudo lvs
LV      VG      Attr   LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
ubuntu-lv ubuntu-vg -wi-ao--- <34.00g
quantha@openstack-aiio:~$ sudo lvextend -l +100%FREE -r /dev/mapper/ubuntu--vg-ubuntu--lv
Size of logical volume ubuntu-vg/ubuntu-lv changed from <34.00 GiB (8703 extents) to <68.00 GiB (17407 extents).
Logical volume ubuntu-vg/ubuntu-lv successfully resized.
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/mapper/ubuntu--vg-ubuntu--lv is mounted on /; on-line resizing required
old_desc_blocks = 5, new_desc_blocks = 9
The filesystem on /dev/mapper/ubuntu--vg-ubuntu--lv is now 17824768 (4k) blocks long.

quantha@openstack-aiio:~$ df -h /\
> ^C
quantha@openstack-aiio:~$
quantha@openstack-aiio:~$ df -h /
Filesystem              Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 67G  6.6G  57G  11% /
quantha@openstack-aiio:~$
```

5. Create cinder-volumes

```

quantha@openstack-ai0:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
8.8.4.4 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
8.8.8.8 via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
10.0.2.2 dev enp0s3 proto dhcp scope link src 10.0.2.15 metric 100
quantha@openstack-ai0:~$ df -h /
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 67G   6.7G   57G   11% /
quantha@openstack-ai0:~$ lsblk
NAME                        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda                         8:0      0    70G  0 disk
├─sda1                      8:1      0     1M  0 part
├─sda2                      8:2      0     2G  0 part /boot
├─sda3                      8:3      0    68G  0 part
└─ubuntu--vg-ubuntu--lv 252:0    0    68G  0 lvm  /
sdb                         8:16     0    30G  0 disk
sr0                         11:0     1 1024M  1 rom
quantha@openstack-ai0:~$ sudo apt install -y lvm2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
lvm2 is already the newest version (2.03.16-3ubuntu3.2).
lvm2 set to manually installed.
The following packages were automatically installed and are no longer required:
  libfwupd2 libgusb2
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
quantha@openstack-ai0:~$ sudo pvcreate /dev/sdb
Physical volume "/dev/sdb" successfully created.
quantha@openstack-ai0:~$ sudo vgcreate cinder-volumes /dev/sdb
Volume group "cinder-volumes" successfully created
quantha@openstack-ai0:~$ sudo pvs
PV          VG          Fmt  Attr  PSize    PFree
/dev/sda3   ubuntu-vg    lvm2  a--   <68.00g    0
/dev/sdb    cinder-volumes lvm2  a--   <30.00g <30.00g
quantha@openstack-ai0:~$ sudo vgs
VG          #PV #LV #SN Attr   VSize    VFree
cinder-volumes 1  0  0 wz--n- <30.00g <30.00g
ubuntu-vg     1  1  0 wz--n- <68.00g    0
quantha@openstack-ai0:~$ sudo lsblk -f
NAME                        FSTYPE      FSVER    LABEL UUID                                FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1                      ext4         1.0              c6bf6a47-0c64-4f66-a77d-8a97ddd859d7    1.7G     5% /boot
├─sda2                      LVM2_member LVM2 001      z59WCx-7kG0-UQYz-cUad-hX3F-25IA-GnwQvw    56.9G    10% /
├─sda3                      ext4         1.0              7838e7cd-bdb3-453e-813c-b9ec77ef5d01    56.9G    10% /
└─ubuntu--vg-ubuntu--lv    LVM2_member LVM2 001      4Li0mr-n21S-qCpc-9oCL-r3a2-00dK-0yNCRa
sdb
sr0

```

6. Update ubuntu & install tools

```
Running kernel seems to be up-to-date.

Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart packagekit.service systemd-journald.service systemd-networkd.service systemd-resolved.service systemd-timesyncd.service systemd-udev.service
udisks2.service upower.service

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service

No containers need to be restarted.

User sessions running outdated binaries:
  quantha @ user manager service: systemd[1333]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
quantha@openstack-ai0:~$ python3 -m venv ~/kolla-venv
quantha@openstack-ai0:~$ source ~/kolla-venv/bin/activate
(kolla-venv) quantha@openstack-ai0:~$ which python
/home/quantha/kolla-venv/bin/python
(kolla-venv) quantha@openstack-ai0:~$ python --version
Python 3.12.3
(kolla-venv) quantha@openstack-ai0:~$ pip --version
pip 24.0 from /home/quantha/kolla-venv/lib/python3.12/site-packages/pip (python 3.12)
(kolla-venv) quantha@openstack-ai0:~$ gentent hosts openstack-ai0
Command 'gentent' not found, did you mean:
  command 'gentent' from deb samba-testsuite (2:4.19.5+dfsg-4ubuntu9.6)
  command 'gentent' from deb libc-bin (2.39-0ubuntu8.7)
Try: sudo apt install <deb name>
(kolla-venv) quantha@openstack-ai0:~$ sudo arping -D -I enp0s3 -c 10.0.2.250
sudo: arping: command not found
(kolla-venv) quantha@openstack-ai0:~$ sudo arping -D -I enp0s3 -c 10.0.2.250
arping: invalid argument: '10.0.2.250'
(kolla-venv) quantha@openstack-ai0:~$ sudo arping -D -I enp0s3 -c 3 10.0.2.250
ARPING 10.0.2.250 from 0.0.0.0 enp0s3
Sent 2 probes (2 broadcast(s))
Received 0 response(s)
(kolla-venv) quantha@openstack-ai0:~$ gentent hosts openstack-ai0
127.0.1.1    openstack-ai0
(kolla-venv) quantha@openstack-ai0:~$ _
```

7. Pip install kolla-ansible

```
(kolla-venv) quantha@openstack-ai0:~$ kolla-ansible --version
kolla-ansible 22.0.0
(kolla-venv) quantha@openstack-ai0:~$ ansible --version
ansible [core 2.20.7]
  config file = None
  configured module search path = ['/home/quantha/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /home/quantha/kolla-venv/lib/python3.12/site-packages/ansible
  ansible collection location = /home/quantha/.ansible/collections:/usr/share/ansible/collections
  executable location = /home/quantha/kolla-venv/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/home/quantha/kolla-venv/bin/python3)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
(kolla-venv) quantha@openstack-ai0:~$ pip show kolla-ansible
Name: kolla-ansible
Version: 22.0.0
Summary: Ansible Deployment of Kolla containers
Home-page: https://docs.openstack.org/kolla-ansible/latest/
Author: OpenStack
Author-email: openstack-discuss@lists.openstack.org
License: Apache License, Version 2.0
Location: /home/quantha/kolla-venv/lib/python3.12/site-packages
Requires: ansible-core, bcrypt, cliff, cryptography, hvac, Jinja2, jmespath, oslo.config, oslo.utils
Required-by:
(kolla-venv) quantha@openstack-ai0:~$
```

8. Copy kolla-ansible files to all-in-one folder

```
(kolla-venv) quantha@openstack-ai0:~$ sudo mkdir -p /etc/kolla
(kolla-venv) quantha@openstack-ai0:~$ sudo chown "%USER": "$USER" /etc/kolla
chown: invalid user: '%USER:quantha'
(kolla-venv) quantha@openstack-ai0:~$ sudo chown "%USER": "$USER" /etc/kolla
(kolla-venv) quantha@openstack-ai0:~$ cp -r "$VIRTUAL_ENV/share/kolla-ansible/etc_examples/kolla/*" /etc/kolla/
(kolla-venv) quantha@openstack-ai0:~$ cp -r "$VIRTUAL_ENV/share/kolla-ansible/inventory/all-in-one*" ~/all-in-one
cp: cannot stat '/home/quantha/kolla-venv/share/kolla-ansible/inventory/all-in-one*': No such file or directory
(kolla-venv) quantha@openstack-ai0:~$ cp -r "$VIRTUAL_ENV/share/kolla-ansible/ansible/inventory/all-in-one*" ~/all-in-one
(kolla-venv) quantha@openstack-ai0:~$ ls -la /etc/kolla
total 48
drwxr-xr-x  2 quantha quantha 4096 Jul 30 15:02 .
drwxr-xr-x 111 root    root   4096 Jul 30 15:01 ..
-rw-rw-r--  1 quantha quantha 31341 Jul 30 15:02 globals.yml
-rw-rw-r--  1 quantha quantha 4534 Jul 30 15:02 passwords.yml
(kolla-venv) quantha@openstack-ai0:~$ ls -la ~/all-in-one
-rw-rw-r--  1 quantha quantha 7693 Jul 30 15:03 /home/quantha/all-in-one
(kolla-venv) quantha@openstack-ai0:~$ _
```

9. Run kolla-ansible bootstrap & prechecks


```

TASK [openstack.kolla.etc_hosts : Ensure hostname does not point to 127.0.1.1 in /etc/hosts] *****
changed: [localhost]

TASK [openstack.kolla.etc_hosts : Generate /etc/hosts for all of the nodes] *****
changed: [localhost]

TASK [openstack.kolla.etc_hosts : Check whether /etc/cloud/cloud.cfg exists] *****
ok: [localhost]

TASK [openstack.kolla.etc_hosts : Disable cloud-init manage_etc_hosts] *****
changed: [localhost]

TASK [openstack.kolla.baremetal : Ensure unprivileged users can use ping] *****
skipping: [localhost]

TASK [openstack.kolla.baremetal : Gather package facts] *****
ok: [localhost]

TASK [openstack.kolla.baremetal : Set firewall default policy] *****
ok: [localhost]

TASK [openstack.kolla.baremetal : Disable firewalld] *****
skipping: [localhost] => (item=firewalld)
skipping: [localhost]

TASK [openstack.kolla.packages : Install packages] *****
ok: [localhost]

TASK [openstack.kolla.packages : Remove packages] *****

```

+ Install docker, dbus

+ prechecks -l --use-test-images

```

skipping: no hosts matched
[WARNING]: Could not match supplied host pattern, ignoring: enable_cloudkitty_True

PLAY [Apply role cloudkitty] *****
skipping: no hosts matched
[WARNING]: Could not match supplied host pattern, ignoring: enable_tacker_True

PLAY [Apply role tacker] *****
skipping: no hosts matched
[WARNING]: Could not match supplied host pattern, ignoring: enable_octavia_True

PLAY [Apply role octavia] *****
skipping: no hosts matched
[WARNING]: Could not match supplied host pattern, ignoring: enable_blazar_True

PLAY [Apply role blazar] *****
skipping: no hosts matched
[WARNING]: Could not match supplied host pattern, ignoring: enable_masakari_True

PLAY [Apply role masakari] *****
skipping: no hosts matched
[WARNING]: Could not match supplied host pattern, ignoring: enable_skyline_True

PLAY [Apply role skyline] *****
skipping: no hosts matched

PLAY RECAP *****
localhost      : ok=116  changed=0    unreachable=0    failed=0    skipped=157    rescued=0    ignored=0

(kolla-venv) quantha@openstack-aio:~$

```

+ pip install python-openstackclient

10. Network

Adapter 1 — NAT

enp0s3

Internet + SSH port forwarding

Adapter 2 — Host-only Management

enp0s8

Ubuntu: 192.168.56.10/24

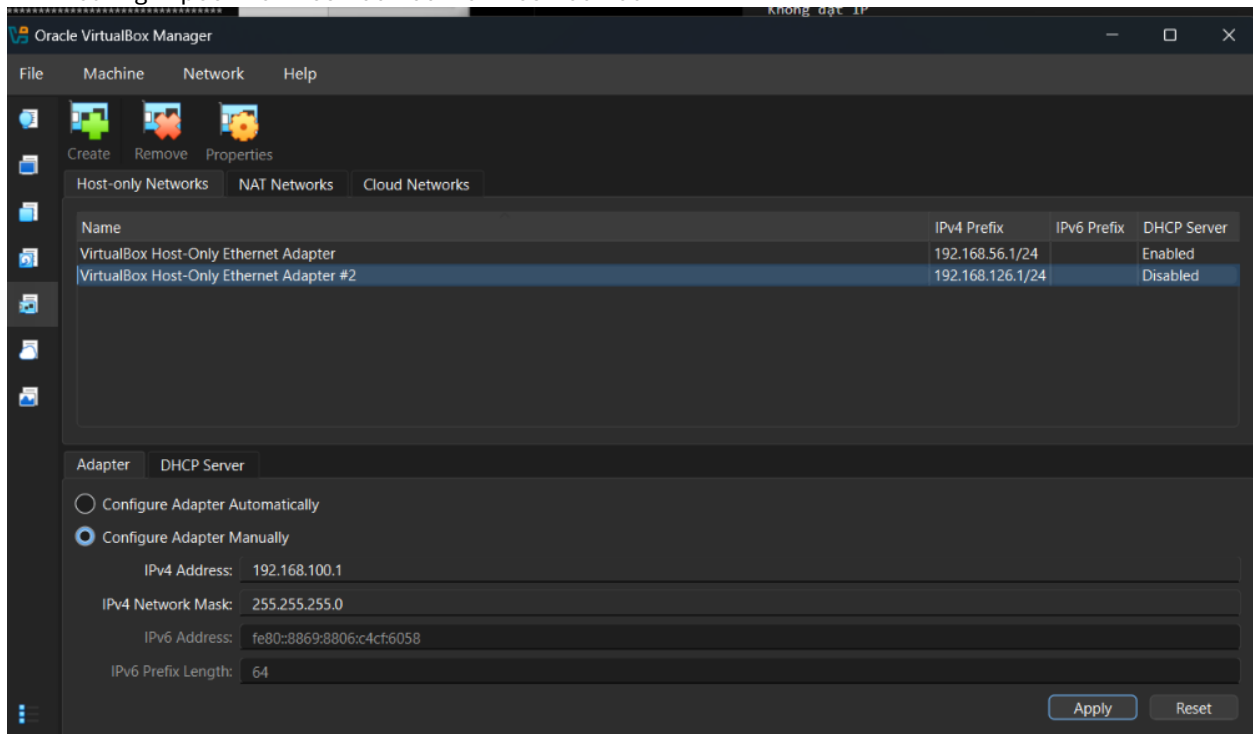
VIP: 192.168.56.20

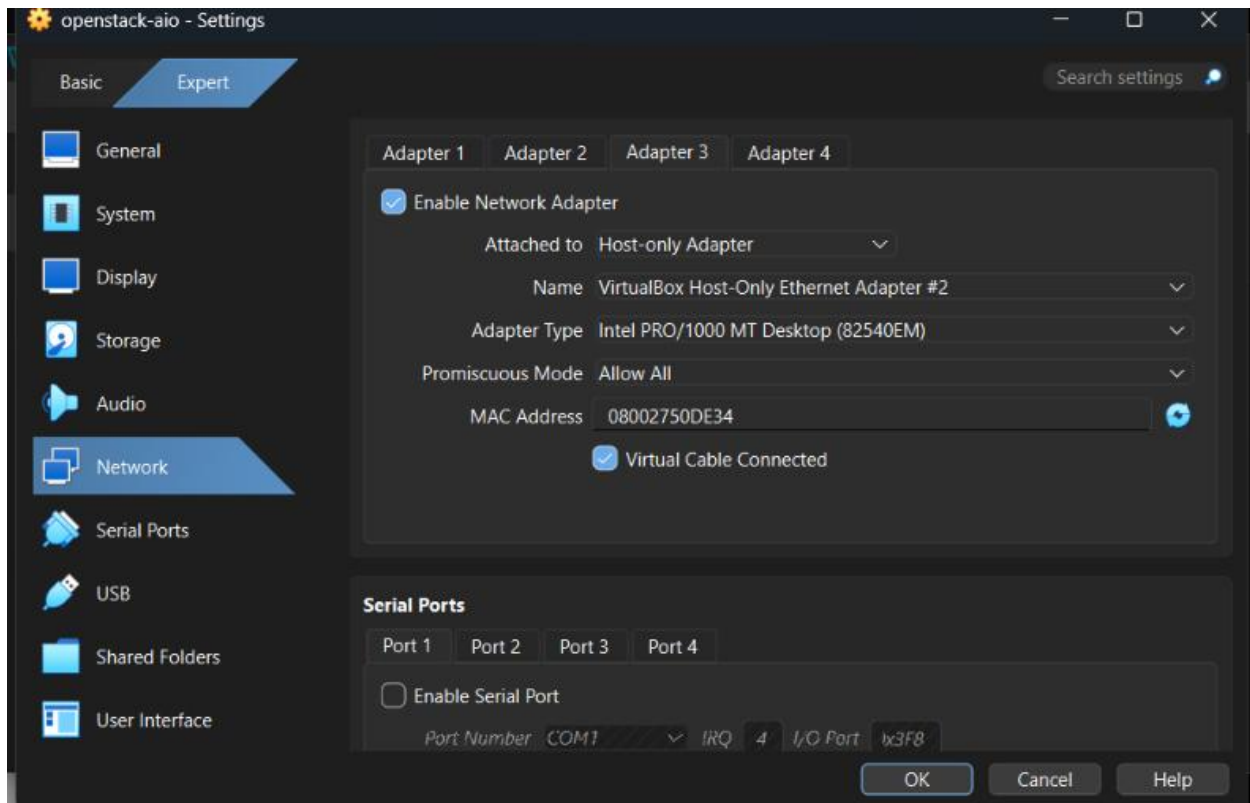
Adapter 3 — Host-only Provider

enp0s9

Không đặt IP

Floating IP pool: 192.168.100.100–192.168.100.200





+ 50-cloud-init.yaml

network:

version: 2

ethernets:

enp0s3:

dhcp4: true

enp0s8:

dhcp4: false

addresses:

- 192.168.56.10/24

optional: true

enp0s9:

dhcp4: false

dhcp6: false

optional: true

+ globals.yml

network_interface: "enp0s8"

neutron_external_interface: "enp0s9"

kolla_internal_vip_address: "192.168.56.20"

11. Check after deployment

Field	Value
expires	2026-08-01T03:20:22+0000
id	gAAAAABqbBR2TqDIA3JJeQ9E4Lwker46tJNrevkDyVsIiJt5UORD9gGJT08GmsADdvmhch9xiEh_8Ra-SLjhf--NMFXCXAkqu5Xi4iCr0x9-NYLvEpTxPiruQpeSBCT_9v7o1gFbR6u-c4NdWtN53IDTrab_HBf-mcF4iEcDlIdqGxDuxYa5weDI
project_id	4d1dc7033bea4bbe93ece9aae64a5808
user_id	6d9923c4224e4e24b484431fd07307ee

```
kolla-venv) quantha@openstack-ai0:~$ openstack service list
```

ID	Name	Type
31c5d93ee8a5f128626d95e7ca44bub	cinder	block-storage
3d59600137184e7e9326a056054bc575	heat	orchestration
6318d9b8c214402eb5ca384062b4dde0	neutron	network
6d49f38515554adeba6270e4eb445d51	glance	image
992d9d23bda54c38a18ec91986c08b6f	nova	compute
cfd8c63ffdb854c07bc9ef44f4a24aaae	heat-cfn	cloudformation
ecab9ea233ef46a984056565c3f444444	keystone	identity
fc954ce864194e48bbe1446f2f1a3a66	placement	placement

```
kolla-venv) quantha@openstack-ai0:~$ openstack compute service list
```

ID	Binary	Host	Zone	Status	State	Updated At
8b65c515-5814-42e6-97d2-807ddc000f0d	nova-scheduler	openstack-ai0	internal	enabled	up	2026-07-31T03:21:47.000000
05c9c567-2e47-422a-a673-6c79a795e4e1	nova-conductor	openstack-ai0	internal	enabled	up	2026-07-31T03:21:40.000000
5f9dd335-36d9-4b9a-91ab-abbddf9d6347	nova-compute	openstack-ai0	nova	enabled	up	2026-07-31T03:21:43.000000

```
kolla-venv) quantha@openstack-ai0:~$ openstack hypervisor list
```

ID	Hypervisor	Hostname	Hypervisor Type	Host IP	State
5370ac7e-6b1a-49ea-8c62-bce5e57a7981	openstack-ai0		QEMU	192.168.56.10	up

```
kolla-venv) quantha@openstack-ai0:~$ openstack network agent list
```

ID	Agent Type	Host	Availability Zone	Alive	State	Binary
47f393ae-9635-465d-a4cd-46147d252918	DHCP agent	openstack-ai0	nova	:-)	UP	neutron-dhcp-agent
a8633239-eec6-485e-8897-6386ce3e8995	L3 agent	openstack-ai0	nova	:-)	UP	neutron-l3-agent
e508e143-f8a5-4d4d-b445-e4ae28e2b379	Metadata agent	openstack-ai0	None	:-)	UP	neutron-metadata-agent
ff52c8d7-5145-4211-bd12-05508d29bcef	Open vSwitch agent	openstack-ai0	None	:-)	UP	neutron-openvswitch-agent

12. Login Openstack

openstack

admin

admin

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Orchestration

Admin

Identity

Project / Compute / Overview

Overview

Limit Summary

Compute

Instances

vCPUs

RAM

Used 0 of 10

Used 0 of 20

Used 0B of 50GB

Volume

Volumes

Volume Snapshots

Volume Storage

Used 0 of 10

Used 0 of 10

Used 0B of 1000GB

Network

Floating IPs

Security Groups

Security Group Rules

Allocated 0 of 50

Used 0 of 10

Used 0 of 100

Networks

Ports

Routers

Used 0 of 100

Used 0 of 500

Used 0 of 10

Usage Summary

Select a period of time to query its usage:

The date should be in YYYY-MM-DD format.

2026-07-30

to

2026-07-31

Submit

Active Instances:

Active RAM:

This Period's vCPU-Hours:

This Period's GB-Hours:

This Period's RAM-Hours:

0

0B

0.00

0.00

0.00

Usage

Download CSV Summary

13. Create & upload cirros image

```
(kolla-venv) quantha@openstack-ai0:~$ openstack image create cirros --file cirros-0.6.3-x86_64-disk.img --disk-format qcow2 --container-format bare --public
```

Field	Value
checksum	87617e24a5e30cb3b87fda8c0764838f
container_format	bare
created_at	2026-07-31T03:26:08Z
disk_format	qcow2
file	/v2/images/068bb3d6-fc22-4710-943d-ee7f66548f1e/file
id	068bb3d6-fc22-4710-943d-ee7f66548f1e
min_disk	0
min_ram	0
name	cirros
owner	4d1dc7033bea4bbe93ece9aae64a5808
properties	os_hash_algo='sha512', os_hash_value='9a9bce0083a00939ec17c11febbfc767aa211aaa54f51e75c5a8b271a9b5637c77205a518b7a2007cb391d23cceb01e0e4e8d64832317151bc85b734b92a7be0', os_hidden='False', owner_specified.openstack.md5='', owner_specified.openstack.object='images/cirros', owner_specified.openstack.sha256='', stores='file'
protected	False
schema	/v2/schemas/image
size	21692416
status	active
tags	
updated_at	2026-07-31T03:26:09Z
virtual_size	117440512
visibility	public

```
(kolla-venv) quantha@openstack-ai0:~$ openstack image list
```

ID	Name	Status
068bb3d6-fc22-4710-943d-ee7f66548f1e	cirros	active

```
(kolla-venv) quantha@openstack-ai0:~$
```

14. Create network & subnet

Private subnet: 10.10.10.0/24

```
(kolla-venv) quantha@openstack-ai0:~$ openstack subnet create private-subnet --network private-net --subnet-range 10.10.10.0/24 --dns-nameserver 8.8.8.8
```

Field	Value
allocation_pools	10.10.10.2-10.10.10.254
cidr	10.10.10.0/24
created_at	2026-07-31T03:30:20Z
description	
dns_nameservers	8.8.8.8
dns_publish_fixed_ip	None
enable_dhcp	True
gateway_ip	10.10.10.1
host_routes	
id	d0922da3-f488-4edf-9e21-df1abf4deebb
ip_version	4
ipv6_address_mode	None
ipv6_ra_mode	None
name	private-subnet
network_id	0a61ab91-8fc6-42ea-ad14-d390edfa476a
project_id	4d1dc7033bea4bbe93ece9aae64a5808
revision_number	0
router:external	False
segment_id	None
service_types	
subnetpool_id	None
tags	
updated_at	2026-07-31T03:30:20Z

```
(kolla-venv) quantha@openstack-ai0:~$ openstack network list
```

ID	Name	Subnets
0a61ab91-8fc6-42ea-ad14-d390edfa476a	private-net	d0922da3-f488-4edf-9e21-df1abf4deebb

```
(kolla-venv) quantha@openstack-ai0:~$ openstack subnet list
```

ID	Name	Network	Subnet
d0922da3-f488-4edf-9e21-df1abf4deebb	private-subnet	0a61ab91-8fc6-42ea-ad14-d390edfa476a	10.10.10.0/24

```
(kolla-venv) quantha@openstack-ai0:~$
```

Public subnet: 192.168.100.0/24

Allocation pool .100 -> .200

Gateway 192.168.100.1

```

quantha@openstack-ai0: ~
port_security_enabled | True
project_id            | 4d1dc7033bea4bbe93ece9aae64a5808
provider:network_type | flat
provider:physical_network | physnet1
provider:segmentation_id | None
pvlan                | None
qos_policy_id         | None
revision_number       | 1
router:external        | External
segments              | None
shared                | False
status                | ACTIVE
subnets              |
tags                  |
updated_at            | 2026-07-31T03:31:55Z
-----
kolla-venv) quantha@openstack-ai0:~$ openstack subnet create public-subnet --network public-net --subnet-range 192.168.100.0/24 --allocation-pool start=192.168.100.100,end=192.168.100.200 --gateway 192.168.100.1 --no-dhcp

Field | Value
-----|-----
allocation_pools | 192.168.100.100-192.168.100.200
cidr          | 192.168.100.0/24
created_at    | 2026-07-31T03:33:00Z
description   |
dns_nameservers |
dns_publish_fixed_ip | None
enable_dhcp    | False
gateway_ip     | 192.168.100.1
host_routes    |
id            | 4347b756-36e9-48b3-b38b-6c7eb272295a
ip_version     | 4
ipv6_address_mode | None
ipv6_ra_mode    | None
name           | public-subnet
network_id      | 79f65a35-5a35-4eb1-a3dd-afc2f8e0519d
project_id      | 4d1dc7033bea4bbe93ece9aae64a5808
revision_number | 0
router:external | True
segment_id      | None
service_types   |
subnetpool_id   | None
tags            |
updated_at      | 2026-07-31T03:33:00Z
-----
kolla-venv) quantha@openstack-ai0:~$

```

```

(kolla-venv) quantha@openstack-ai0:~$ openstack network list

+-----+-----+-----+
| ID | Name | Subnets |
+-----+-----+-----+
| 0a61ab91-8fc6-42ea-ad14-d390edfa476a | private-net | d0922da3-f488-4edf-9e21-df1abf4deebb |
| 79f65a35-5a35-4eb1-a3dd-afc2f8e0519d | public-net | 4347b756-36e9-48b3-b38b-6c7eb272295a |
+-----+-----+-----+

(kolla-venv) quantha@openstack-ai0:~$ openstack subnet list

+-----+-----+-----+-----+
| ID | Name | Network | Subnet |
+-----+-----+-----+-----+
| 4347b756-36e9-48b3-b38b-6c7eb272295a | public-subnet | 79f65a35-5a35-4eb1-a3dd-afc2f8e0519d | 192.168.100.0/24 |
| d0922da3-f488-4edf-9e21-df1abf4deebb | private-subnet | 0a61ab91-8fc6-42ea-ad14-d390edfa476a | 10.10.10.0/24 |
+-----+-----+-----+-----+

(kolla-venv) quantha@openstack-ai0:~$

```

15. Create router

```

(kolla-venv) quantha@openstack-ai0:~$ openstack router add subnet lab-router private-subnet
(kolla-venv) quantha@openstack-ai0:~$ openstack router set lab-router --external-gateway public-net
(kolla-venv) quantha@openstack-ai0:~$ openstack router show lab-router

+-----+-----+
| Field | Value |
+-----+-----+
| admin_state_up | UP |
| availability_zone_hints | nova |
| availability_zones | 2026-07-31T03:35:47Z |
| created_at | 2026-07-31T03:35:47Z |
| description | None |
| distributed | False |
| enable_ndp_proxy | None |
| external_gateway_info | {"network_id": "79f65a35-5a35-4eb1-a3dd-afc2f8e0519d", "external_fixed_ips": [{"subnet_id": "4347b756-36e9-48b3-b38b-6c7eb272295a", "ip_address": "192.168.100.101"}], "enable_snat": true} |
| flavor_id | None |
| ha | False |
| id | e49862c8-5320-47e8-be14-cbf1bbd384d9 |
| interfaces_info | [{"port_id": "bd9b1c54-5264-4012-a343-15a01209e048", "ip_address": "10.10.10.1", "subnet_id": "d0922da3-f488-4edf-9e21-df1abf4deebb"}] |
| name | lab-router |
| project_id | 4d1dc7033bea4bbe93ece9aae64a5808 |
| revision_number | 3 |
| routes |
| status | ACTIVE |
| tags |
| updated_at | 2026-07-31T03:36:22Z |
+-----+-----+

(kolla-venv) quantha@openstack-ai0:~$

```

16. Security Group

Icmp

Ssh port 22

```
(kolla-venv) quantha@openstack-ai0:~$ openstack security group rule create --protocol icmp default
+-----+-----+
| Field | Value |
+-----+-----+
| belongs_to_default_sg | True |
| created_at | 2026-07-31T03:38:05Z |
| description | |
| direction | ingress |
| ether_type | IPv4 |
| id | c6ea2b4a-5519-4216-97b6-7437730428d3 |
| normalized_cidr | 0.0.0.0/0 |
| port_range_max | None |
| port_range_min | None |
| project_id | 4d1dc7033bea4bbe93ece9aae64a5808 |
| protocol | icmp |
| remote_address_group_id | None |
| remote_group_id | None |
| remote_ip_prefix | 0.0.0.0/0 |
| revision_number | 0 |
| security_group_id | e03ef713-f3f0-48b8-9ba7-c9668531b411 |
| updated_at | 2026-07-31T03:38:05Z |
+-----+-----+
(kolla-venv) quantha@openstack-ai0:~$ openstack security group rule create --protocol tcp --dst-port 22 default
+-----+-----+
| Field | Value |
+-----+-----+
| belongs_to_default_sg | True |
| created_at | 2026-07-31T03:38:22Z |
| description | |
| direction | ingress |
| ether_type | IPv4 |
| id | 46192a31-949d-4737-b229-03eab4f7da00 |
| normalized_cidr | 0.0.0.0/0 |
| port_range_max | 22 |
| port_range_min | 22 |
| project_id | 4d1dc7033bea4bbe93ece9aae64a5808 |
| protocol | tcp |
| remote_address_group_id | None |
| remote_group_id | None |
| remote_ip_prefix | 0.0.0.0/0 |
| revision_number | 0 |
| security_group_id | e03ef713-f3f0-48b8-9ba7-c9668531b411 |
| updated_at | 2026-07-31T03:38:22Z |
+-----+-----+
(kolla-venv) quantha@openstack-ai0:~$
```

```
(kolla-venv) quantha@openstack-ai0:~$ openstack security group rule list default
+-----+-----+-----+-----+-----+-----+-----+-----+
| ID | IP Protocol | Ethertype | IP Range | Port Range | Direction | Remote Security Group | Remote Address Group |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 46192a31-949d-4737-b229-03eab4f7da00 | tcp | IPv4 | 0.0.0.0/0 | 22:22 | ingress | None | None |
| 6eb6f92b-2ee3-4644-9602-7e0ea1bf08ab | None | IPv4 | 0.0.0.0/0 | | egress | None | None |
| 71ca14a9-59c7-42b5-ba4f-d86218608243 | None | IPv6 | ::/0 | | ingress | e03ef713-f3f0-48b8-9ba7-c9668531b411 | None |
| 78ade065-d5a7-4606-9a01-1f533f60f791 | None | IPv4 | 0.0.0.0/0 | | ingress | e03ef713-f3f0-48b8-9ba7-c9668531b411 | None |
| b98351b1-446c-4223-9e55-f4860eff947e | None | IPv6 | ::/0 | | egress | None | None |
| c6ea2b4a-5519-4216-97b6-7437730428d3 | icmp | IPv4 | 0.0.0.0/0 | | ingress | None | None |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

Private interface: 10.10.10.1

External gateway: 192.168.100.101

External network: public-net

10.10.10.1 là cổng trong của router trên private-subnet. VM trong mạng 10.10.10.0/24 sẽ dùng địa chỉ này làm default gateway.

192.168.100.101 là địa chỉ Neutron cấp cho cổng ngoài của lab-router. Địa chỉ này được lấy tự động từ allocation pool:

192.168.100.100–192.168.100.200

Nó không phải gateway vật lý. Gateway vật lý của public subnet vẫn là:

192.168.100.1

Trong lab:

VM: 10.10.10.x

|



lab-router internal: 10.10.10.1

| NAT



lab-router external: 192.168.100.101

|



VirtualBox Host-only gateway/Windows: 192.168.100.1

public-net là provider network nối qua:

physnet1 → br-ex → enp0s9 → VirtualBox Host-only network

Việc Neutron chọn .101 thay vì .100 là bình thường; địa chỉ external gateway port được cấp động từ allocation pool và không cần cố định ở địa chỉ đầu tiên.

17. Key pair for ssh

```
(kolla-venv) quantha@openstack-ai0:~$ ssh-keygen -t ed25519 -f ~/.ssh/openstack_lab
Generating public/private ed25519 key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/quantha/.ssh/openstack_lab
Your public key has been saved in /home/quantha/.ssh/openstack_lab.pub
The key fingerprint is:
SHA256:0GyzVSRqH6NX7EV4HhG0tQe55c8kMdZ4KK+nJ9009iY quantha@openstack-ai0
The key's randomart image is:
+--[ED25519 256]--+
  .,=+=
  .+B.X o
  o oo+%=
  o +.*o.* .
  B S .o =
  . * . . *
  . + + o
  . o o E o
  o o
+-----[SHA256]-----+
(kolla-venv) quantha@openstack-ai0:~$ openstack keypair create --public-key ~/.ssh/openstack_lab.pub lab-key
```

Field	Value
created_at	None
fingerprint	08:db:29:50:75:9e:d3:c4:9f:6b:ce:69:83:f4:c7:c6
id	lab-key
is_deleted	None
name	lab-key
type	ssh
user_id	6d9923c4224e4e24b484431fd07307ee

```
(kolla-venv) quantha@openstack-ai0:~$ openstack keypair list
```

Name	Fingerprint	Type
lab-key	08:db:29:50:75:9e:d3:c4:9f:6b:ce:69:83:f4:c7:c6	ssh

```
(kolla-venv) quantha@openstack-ai0:~$
```

18. Boot VM cirrOS

```
OS-EXT-SRV-ATTR:root_device_name      None
OS-EXT-SRV-ATTR:user_data              None
OS-EXT-STS:power_state                 N/A
OS-EXT-STS:task_state                  scheduling
OS-EXT-STS:vm_state                    building
OS-SRV-USG:launched_at                 None
OS-SRV-USG:terminated_at               None
accessIPv4                              None
accessIPv6                              None
addresses                               N/A
adminPass                               7G2k38inxEbD
config_drive                           None
created                                2026-07-31T03:42:41Z
description                             None
flavor                                  description=, disk='5', ephemeral='0', , id='ml.small', is_disabled=, is_public='True', location=,
name='ml.small', original_name='ml.small', ram='1024', rxtx_factor=, swap='0', vcpus='1'
hostId                                  None
host_status                             None
id                                       3cd973d3-377c-4a62-a105-c22905da284e
image                                   cirros (068bb3d6-fc22-4710-943d-ee7f66548f1e)
key_name                                lab-key
locked                                  None
locked_reason                           None
name                                    test-vm
pinned_availability_zone                 None
progress                                None
project_id                              4d1dc7033bea4bbe93ece9aae64a5808
properties                              None
scheduler_hints                         None
security_groups                         name='default'
server_groups                           None
status                                  BUILD
tags                                    None
trusted_image_certificates              None
updated                                2026-07-31T03:42:42Z
user_id                                 6d9923c4224e4e24b484431fd07307ee
volumes_attached                        None
(kolla-venv) quantha@openstack-ai0:~$ watch -n 2 openstack server list
(kolla-venv) quantha@openstack-ai0:~$ openstack server list
```

ID	Name	Status	Networks	Image	Flavor
3cd973d3-377c-4a62-a105-c22905da284e	test-vm	ACTIVE	private-net=10.10.10.59	cirros	ml.small

```
(kolla-venv) quantha@openstack-ai0:~$
```

19. Create floating ip

```
kolla-venv) quantha@openstack-aio:~$ openstack floating ip create public-net
```

Field	Value
created_at	2026-07-31T03:44:31Z
description	
dns_domain	None
dns_name	None
fixed_ip_address	None
floating_ip_address	192.168.100.166
floating_network_id	79f65a35-5a35-4eb1-a3dd-afc2f8e0519d
id	cffc0ec7-3b30-4a82-95f1-d1b388442c80
name	192.168.100.166
port_details	None
port_id	None
project_id	4d1dc7033bea4bbe93ece9aae64a5808
qos_policy_id	None
revision_number	0
router_id	None
status	DOWN
subnet_id	None
tags	[]
updated_at	2026-07-31T03:44:31Z

Address: 192.168.100.166

openstack server add floating ip test-vm 192.168.100.166

```
(kolla-venv) quantha@openstack-aio:~$ openstack server list
```

ID	Name	Status	Networks	Image	Flavor
3cd973d3-377c-4a62-a105-c22905da284e	test-vm	ACTIVE	private-net=10.10.10.59, 192.168.100.166	cirros	m1.small

Test ping to local powershell:

```
PS C:\WINDOWS\system32> ping 192.168.100.166

Pinging 192.168.100.166 with 32 bytes of data:
Reply from 192.168.100.166: bytes=32 time=3ms TTL=63
Reply from 192.168.100.166: bytes=32 time=1ms TTL=63
Reply from 192.168.100.166: bytes=32 time<1ms TTL=63
Reply from 192.168.100.166: bytes=32 time=5ms TTL=63

Ping statistics for 192.168.100.166:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 5ms, Average = 2ms
PS C:\WINDOWS\system32>
```

Ssh:

```
PS C:\WINDOWS\system32> scp quantha@192.168.56.10:/home/quantha/.ssh/openstack_lab C:\Users\ADMIN\.ssh\openstack_lab
quantha@192.168.56.10's password:
openstack_lab 100% 411 200.7KB/s 00:00
PS C:\WINDOWS\system32> ssh -i C:\Users\ADMIN\.ssh\openstack_lab cirros@192.168.100.166
$
$
$
```

20. Cinder Volume

```
(kolla-venv) quantha@openstack-aiio:~$ openstack volume create --size 1 data-volume
```

Field	Value
attachments	[]
availability_zone	nova
backup_id	None
bootable	False
cluster_name	None
consumes_quota	True
created_at	2026-07-31T03:50:45.885147
description	None
encrypted	False
group_id	None
id	42d2eab9-16ec-4357-ac6a-0aa007572340
multiattach	False
name	data-volume
os-vol-host-attr:host	None
os-vol-mig-status-attr:migstat	None
os-vol-mig-status-attr:name_id	None
os-vol-tenant-attr:tenant_id	None
properties	
provider_id	None
replication_status	None
service_uuid	None
shared_targets	True
size	1
snapshot_id	None
source_volid	None
status	creating
type	__DEFAULT__
updated_at	None
user_id	6d9923c4224e4e24b484431fd07307ee
volume_type_id	e9a05556-6191-495a-9f3e-bc937b65d811

```
(kolla-venv) quantha@openstack-aiio:~$ openstack volume list
```

ID	Name	Status	Size	Attached to
42d2eab9-16ec-4357-ac6a-0aa007572340	data-volume	available	1	

```
(kolla-venv) quantha@openstack-aiio:~$
```

openstack server add volume test-vm data-volume

```
(kolla-venv) quantha@openstack-aiio:~$ openstack server add volume test-vm data-volume
```

Field	Value
ID	42d2eab9-16ec-4357-ac6a-0aa007572340
Server ID	3cd973d3-377c-4a62-a105-c22905da284e
Volume ID	42d2eab9-16ec-4357-ac6a-0aa007572340
Device	/dev/vdb
Tag	None
Delete On Termination	False

```
(kolla-venv) quantha@openstack-aiio:~$ openstack volume list
```

ID	Name	Status	Size	Attached to
42d2eab9-16ec-4357-ac6a-0aa007572340	data-volume	in-use	1	Attached to test-vm on /dev/vdb

Tao filesystem ext4 & mount volume de test persistent tren cirros

```

$ sudo mkfs.ext4 /dev/vdb
mke2fs 1.46.5 (30-Dec-2021)
Discarding device blocks: done
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 680d338b-2006-4df3-8bf3-0958461714fe
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

$ sudo mkdir -p /mnt/data
$ sudo mount /dev/vdb /mnt/data
$ echo "OpenStack Cinder persistent test" | sudo tee /mnt/data/test.txt
OpenStack Cinder persistent test
$ cat /mnt/data/test.txt
cat: can't open '/mnt/data/test.txt': Permission denied
$ sudo cat /mnt/data/test.txt
OpenStack Cinder persistent test
$

```

Data-volume still available after umount

```

(kolla-venv) quantha@openstack-aioc:~$ openstack server remove volume test-vm data-volume
(kolla-venv) quantha@openstack-aioc:~$ openstack volume list
+-----+-----+-----+-----+-----+
| ID | Name | Status | Size | Attached to |
+-----+-----+-----+-----+-----+
| 42d2eab9-16ec-4357-ac6a-0aa007572340 | data-volume | available | 1 | |
+-----+-----+-----+-----+-----+
(kolla-venv) quantha@openstack-aioc:~$

```

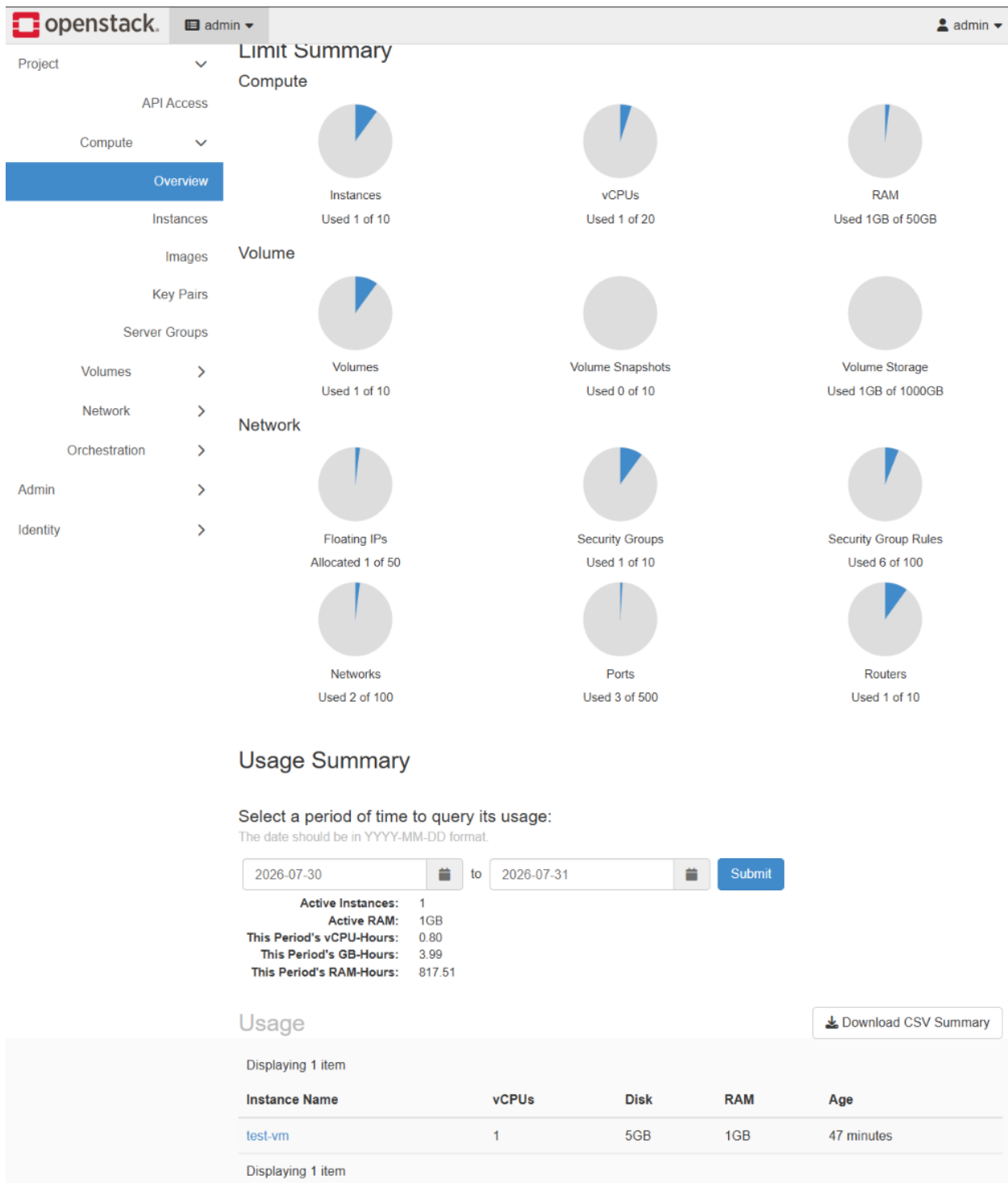
Data persists

```

S C:\WINDOWS\system32> ssh -i C:\Users\ADMIN\.ssh\openstack_lab cirros@192.168.100.166
sudo mount /dev/vdb /mnt/data
sudo cat /mnt/data/test.txt
penStack Cinder persistent test

```

OpenStack Dashboard



```

GNU nano 7.2 /home/quantha/heat-test.yaml
heat_template_version: 2021-04-16

Description: Heat test stack creating a network and subnet

resources:
  heat_test_network:
    type: OS::Neutron::Net
    properties:
      name: heat-test-net

  heat_test_subnet:
    type: OS::Neutron::Subnet
    properties:
      name: heat-test-subnet
      network: { get_resource: heat_test_network }
      cidr: 10.20.30.0/24
      gateway_ip: 10.20.30.1
      enable_dhcp: true

outputs:
  network_id:
    description: ID of the created network
    value: { get_resource: heat_test_network }

  subnet_id:
    description: ID of the created subnet
    value: { get_resource: heat_test_subnet }

```

Openstack stack create -t ~/heat-test.yaml heat-test-stack

```

(kolla-venv) quantha@openstack-aio:~$ openstack stack create -t ~/heat-test.yaml heat-test-stack

```

Field	Value
id	853faf32-2c47-4e45-bd4b-178d65b6db89
stack_name	heat-test-stack
description	Heat test stack creating a network and subnet
creation_time	2026-07-31T07:11:17Z
updated_time	None
stack_status	CREATE_COMPLETE
stack_status_reason	Stack CREATE completed successfully

```

(kolla-venv) quantha@openstack-aio:~$ nano ~/heat-test.yaml
(kolla-venv) quantha@openstack-aio:~$ openstack stack list

```

ID	Stack Name	Project	Stack Status	Creation Time	Updated Time
853faf32-2c47-4e45-bd4b-178d65b6db89	heat-test-stack	4d1dc7033bea4bbe93ece9aae64a5808	CREATE_COMPLETE	2026-07-31T07:11:17Z	None

```

(kolla-venv) quantha@openstack-aio:~$ openstack stack show heat-test-stack

```

Field	Value
id	853faf32-2c47-4e45-bd4b-178d65b6db89
stack_name	heat-test-stack
description	Heat test stack creating a network and subnet
creation_time	2026-07-31T07:11:17Z
updated_time	None
stack_status	CREATE_COMPLETE
stack_status_reason	Stack CREATE completed successfully
parameters	OS::project_id: 4d1dc7033bea4bbe93ece9aae64a5808 OS::stack_id: 853faf32-2c47-4e45-bd4b-178d65b6db89 OS::stack_name: heat-test-stack
outputs	- description: ID of the created subnet output_key: subnet_id output_value: a8983a6d-82b2-4024-a36b-e9f7daadd892 - description: ID of the created network output_key: network_id output_value: b3f2ef81-7212-4eda-b6ad-70de6d5a4e54
links	- href: http://192.168.56.20:8004/v1/4d1dc7033bea4bbe93ece9aae64a5808/stacks/heat-test-stack/853faf32-2c47-4e45-bd4b-178d65b6db89 rel: self
deletion_time	None
notification_topics	[]
capabilities	[]
disable_rollback	True
timeout_mins	None
stack_owner	admin
parent	None
stack_user_project_id	ec3c11805720490ea9618d285c7calae
tags	[]

- Network heat-test-net được tạo tự động bởi Heat, được show trong Neutron với status ACTIVE. Network có bật port security, có MY 1450 và liên kết với subnet do cùng stack tạo ra. → Heat tạo ra stack và giao tiếp với Neutron API → tạo network resources
- Heat Orchestration được test bằng HOT Template tạo tự động một Neutron network và một subnet. Network heat-test-net ACTIVE trong Neutron, heat-test-subnet được tạo với CIDR 10.20.30.0/24, gateway .1, DHCP enabled và allocation pool từ .2 đến .254 → Heat API, Heat Engine, Keystone Authentication và Heat gọi Neutron API works.
- Placement đã được kiểm thử trực tiếp bằng kiểm tra resource provider, inventory và allocation của instance. Resource provider openstack-aio quản lý tài nguyên CPU, RAM, disk và compute host. Instance test-vm được Placement record sử dụng 1 vCPU, 1024 MB RAM và 5 GB disk – m1.small.
- Glance được test bằng xem chi tiết image cirros. Image active, định dạng qcow2, visibility public và được lưu trên backend file. Checksum MD5 và hash SHA-512. Image để Nova tạo instance test-vm